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Chapter 13 Stage 13 of 15 High

Test and Assembly

Testing finds bad chips before they go into an expensive package. Japan's Advantest took about 65 percent of 2025 spending on chip testers, by its own estimate.

Specimen 13 · Test and Assembly

A vertical probe card over a wafer. It is called vertical because its needles point straight down at the wafer. The card is round, about 305 mm across, a little wider than the 300 mm wafer beneath it. Thousands of needles under it land on four dies at once, and those four dies are picked out in blue-violet. The gap between card and wafer is exaggerated. The real gap is a millimeter or two.

In plain terms

Before a chip can be sold, a machine called a tester has to check that it works. While the chips are still on the wafer, thousands of needles press onto small metal pads on each one. The tester sends signals through the needles and reads what comes back, and any chip that fails goes no further. A modern AI accelerator is a dozen expensive pieces bonded into one package, and if one piece is bad, all of them are thrown away. So the industry now tests chips earlier, at higher temperatures and for longer, to make weak ones fail during testing before they reach a finished product. Two firms, Advantest in Japan and Teradyne in the United States, sell most of the testers.

In short

In a TrendForce ranking cited by CSET, four of the ten largest chip assembly and test contractors by 2024 revenue are headquartered in China[9]. Georgetown's CSET, analyzing TechInsights data, finds that between 2019 and 2024 Chinese toolmakers gained 5 percentage points of world market share in test tools[9]. Across assembly, test and packaging tools, they still lag significantly behind foreign suppliers in market share[9]. Advantest says that switching to another tester maker means rebuilding a customer's whole test setup, at significant extra cost[8].

8 **Investors Guide July 29, 2026.** Advantest, 30 July 2026, p. 12. advantest.com

9 **Inside Beijing's Chipmaking Offensive.** Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

Chokepoint card

Test and Assembly, stage 13 of 15

Concentration

High

Four to six firms could supply this stage today, and the top two take most of the revenue.

Substitutability

Moderate

Rated on test tools, where Advantest held 62.7 percent of the market in 2025 and Teradyne 21.8 percent, by CSET's analysis of TechInsights data [1].

Price or market size

About $9.0 billion

market for chip testers in 2025, by Advantest's estimate [2]

Who leads	JP	**Advantest**	About 65% of the tester market in 2025, by its own estimate [2]
	US	**Teradyne**	$2.52bn of semiconductor test revenue in 2025, service included [3]
	TW	**ASE**	Ran 8,348 testers in Q2 2026, up from 6,797 a year earlier [4]
	US	**Amkor**	$6.71bn of 2025 revenue [5]
	JP	**Disco**	87% of the $1.9bn market for dicing saws and grinders in 2025, on CSET's analysis of TechInsights data [6]
Where it is made	TW	**Taiwan**	Taiwanese firms such as ASE and Powertech lead outsourced assembly and test
	JP	**Japan**	Advantest testers, Accretech and Tokyo Electron probers, Disco and Accretech dicing and grinding
	US	**United States**	Test equipment from Teradyne, FormFactor, Cohu and Aehr, and Amkor's planned Arizona packaging plant
	CN	**China**	Four of the top ten assembly and test firms by 2024 revenue, in a TrendForce ranking cited by CSET
	MY	**Malaysia**	Malaysia said in July 2026 that it ships 13% of the world's packaged chips
Why substitution is possible			Customers such as ASE already buy testers from both firms [7]. The barrier is cost, because switching means rebuilding a customer's whole test setup, Advantest says [8].

Where China stands

Georgetown's CSET, analyzing TechInsights data, finds that between 2019 and 2024 Chinese toolmakers gained 5 percentage points of world market share in test tools and 4 points in fabrication tools built for advanced packaging [9]. But they lost share in bonding and other conventional assembly and packaging tools [9].

Where the US stands

American firms held 23.5 percent of the market for test tools in 2025 and Japanese firms 63.0 percent, by CSET's analysis of TechInsights data [1]. Amkor raised its planned investment in an Arizona packaging and test campus to $7 billion in October 2025 [10]. In June 2026 it announced a long-term partnership with TSMC, which will buy packaging and test services from it in Arizona [11].

Turning silicon into a finished, packaged chip takes 1,000 to 1,500 steps, depending on how complex the chip is, the Semiconductor Industry

1 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech

2 Advantest, presentation notes for the FY2025 (year ended 31 March 2026) results... Advantest, 27 April 2026, p. 13. advantest.com

3 TERADYNE, INC, Form 10-K annual report for the period ended 2025-12-31 (10-K). U.S. Securities and Exchange Commission, 19 February 2026. sec.gov

4 ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for... U.S. Securities and Exchange Commission, 30 July 2026. sec.gov

5 AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-02-09... U.S. Securities and Exchange Commission, 9 February 2026. sec.gov

6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

7 ASE Technology Holding Co., Ltd., Form 20-F annual report for the period ended... U.S. Securities and Exchange Commission, 1 April 2026. sec.gov

8 Investors Guide July 29, 2026. Advantest, 30 July 2026, p. 12. advantest.com

9 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

10 Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test... Amkor Technology, 6 October 2025. ir.amkor.com

11 TSMC and Amkor Technology Announce Long Term Partnership to Accelerate Advanced... Amkor Technology, 16 June 2026. ir.amkor.com

Association estimated in 2025[12]. Of all those steps, only test puts electrical signals through the chip, Advantest's investor guide of July 2026 says[13]. The earlier stages of chipmaking make dies, the individual chips on a wafer. The back end of the industry, assembly and test, checks which of them work and puts the good ones into packages.

01

How it works

Figure 13.1 · Explainer, 23 seconds

How a chip is tested



1. **The wafer.** Earlier stages make the chips. Before any of them can be sold, a machine tests each one, starting while the chips are still on the wafer.
2. **Needles.** Thousands of needles on a probe card press onto one chip. They touch small metal pads on its surface.
3. **Questions and answers.** Through the needles, the tester sends signals into the chip and reads what comes back. It checks every answer.
4. **Chip by chip.** The wafer moves under the needles, chip by chip. The tester marks any chip that answers wrong, and that chip goes no further.
5. **Into the package.** Only chips that pass go on to be packaged. A modern accelerator bonds together a dozen expensive pieces, and one bad piece throws away all of them, so each piece is tested before it goes in.

Note Simplified. Nine needles stand for thousands. A real prober, the machine that moves the wafer under the card, holds the card still and raises the wafer to meet the needles. The failed chips are placed for illustration and do not reflect a real yield. The package follows chapter 11: two logic chips and eight memory stacks on an interposer and a package base.

Each round of test catches failures the round before it cannot see.

- **Wafer sort.** Thousands of needles land on each die while it is still on the wafer. A tester sends in test patterns and checks the answers, and dies that fail go no further.

12 Testimony of David Isaacs, Semiconductor Industry Association, hearing on… U.S. Senate Committee on Finance, 14 May 2025, p. 2. finance.senate.gov

13 Investors Guide July 29, 2026. Advantest, 30 July 2026, p. 10. advantest.com

- **Burn-in.** The part runs hot and at raised voltage for minutes to days, on the wafer or in the package, so that early-life failures happen in the factory, before the part ships[14].
- **Final test.** The tester checks the packaged part again, because packaging adds failures of its own.
- **System-level test.** The finished part sits in a socket and runs something close to a real workload, because some failures appear only when everything runs at once.

As the parts in a package grow more valuable, finding a bad die before packaging matters more[15]. Aehr, an American maker of burn-in equipment, sells wafer-level burn-in systems that look for early-life and hidden failures before dies are cut from the wafer[15]. Catching those failures early lowers the risk of packaging high-bandwidth memory and an expensive substrate with a processor that may later fail[15]. Aehr says its lead AI customer is moving all burn-in of that customer's AI accelerators from the system level, after packaging, to the wafer[16]. The FOX-XP systems that customer ordered from Aehr in August 2026 each burn in nine 300 mm wafers at a time[15].

02

Variants and trade-offs

Wafer sort and known-good die

The hard part of wafer sort is the probe card, the circuit board that holds the needles. For a large logic die every needle must sit in the same plane, survive repeated touchdowns without drifting, and do it hot. Each card is built for one chip design, so every new design needs a new card and a new qualification. FormFactor, an American probe card maker, says its cards work at high enough frequencies to compete well in testing high-bandwidth memory, a stack of eight, twelve or sixteen memory dies[17].

Final test, burn-in and system-level test

The bigger the package, the more one bad die can cost. TSMC told investors in July 2026 that its CoWoS roadmap includes packages larger than 14 times the area a lithography machine can print in one shot[18]. A die that fails after assembly wastes the memory and substrate joined to it[15].

14 AEHR TEST SYSTEMS, Form 10-K annual report for the period ended 2026-05-29. U.S. Securities and Exchange Commission, 27 July 2026. sec.gov

15 Aehr Receives $22 Million Follow-On Order for AI Processor Wafer-Level Burn-In... Aehr Test Systems, 12 August 2026. aehr.com

16 Aehr Test Systems Reports Fiscal 2026 Fourth Quarter and Full Year Financial... Aehr Test Systems, 14 July 2026. aehr.com

17 FORMFACTOR INC, Form 10-K annual report for the period ended 2025-12-27 (10-K). U.S. Securities and Exchange Commission, 20 February 2026. sec.gov

18 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... LSEG StreetEvents, 16 July 2026, p. 16. investor.tsmc.com

Figure 13.2

Advantest put its tester market share at about 65 percent in 2025

Automated test equipment: share of the ~$9.0bn tester market, 2025



Source Advantest FY2025 results briefing, April 2026 [2]

03

Who makes it

Advantest puts its own share at about 65 percent of an estimated $9.0 billion tester market in 2025[2]. Its share of system-on-chip (SoC) testers, the machines that test processors and other non-memory chips, rose about ten points in a year to 66 percent[2]. It held about 60 percent of memory test[2].

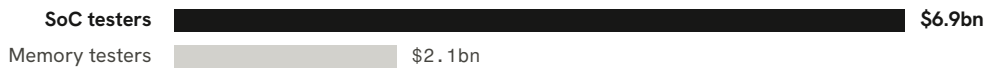
Teradyne, Advantest's main rival, took $2.52 billion of semiconductor test revenue in 2025, service included[3]. Of that, $1.89 billion came from SoC testers and $505 million from memory testers[3]. Teradyne's total revenue grew 13 percent to $3.19 billion in 2025, with fourth-quarter revenue up 44 percent on AI demand[19].

Georgetown University's Center for Security and Emerging Technology (CSET) makes a second estimate, from data by the research firm TechInsights[1]. It gives Advantest 62.7 percent of a $9.1 billion market for test tools in 2025 and Teradyne 21.8 percent[1]. Japanese firms held 63.0 percent of that market and American firms 23.5 percent[1].

Figure 13.3

SoC testers took an estimated $6.9 billion in 2025, memory $2.1 billion

Where the tester money went in calendar 2025



Source Advantest FY2025 results briefing, April 2026 [2]

The rest of the test and assembly equipment market splits into niches, each with its own suppliers.

- **Probe cards.** FormFactor took $785.0 million of revenue in fiscal 2025, most of it from probe cards[17]. Technoprobe's revenue grew 15.7 percent to EUR

1 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech

2 Advantest, presentation notes for the FY2025 (year ended 31 March 2026) results... Advantest, 27 April 2026, p. 13. advantest.com

3 TERADYNE, INC, Form 10-K annual report for the period ended 2025-12-31 (10-K). U.S. Securities and Exchange Commission, 19 February 2026. sec.gov

17 FORMFACTOR INC, Form 10-K annual report for the period ended 2025-12-27 (10-K). U.S. Securities and Exchange Commission, 20 February 2026. sec.gov

19 Teradyne Reports Fourth Quarter and Full Year 2025 Results. Teradyne, 2 February 2026. investors.teradyne.com

628.4 million in 2025, about 38 percent of it from AI[20]. It aims to double its manufacturing capacity by the end of 2027[20].

- **Probers**, the machines that move each wafer under the probe card and keep it at a set temperature. Tokyo Seimitsu, which sells under the Accretech name, says its own research gives it the largest global share in probers[21]. Tokyo Electron makes probers too[22]. CSET's analysis of TechInsights data counts handlers and probers as one market, worth $2.9 billion in 2025, and gives Tokyo Electron 19.9 percent of it and Accretech 17.1 percent[23].
- **Handlers and burn-in**, the machines that feed packaged parts into a tester and hold them at a set temperature, and the systems that run burn-in. Cohu took $453.0 million of revenue in fiscal 2025[24]. Aehr's revenue was $50.0 million in fiscal 2026, but it booked a record $60.7 million of orders in the fourth quarter alone[16]. It expects revenue of $130 million to $150 million in fiscal 2027[16].
- **Dicing and grinding.** Dicing saws cut wafers into dies, and grinders thin them. CSET's analysis of TechInsights data found that in 2025 Disco held 87 percent and Accretech 8.1 percent of the $1.9 billion market for these machines[6]. Panel-level packaging builds packages on large panels in place of 300 mm wafers, to make more of them at once[25]. In December 2025 Disco announced a saw that cuts such panels, as large as 400 by 400 mm, into individual packages[25].
- **Bonding.** Kulicke and Soffa took $654.1 million of revenue in fiscal 2025[26]. Besi's second-quarter 2026 revenue rose 68.7 percent to EUR 249.9 million on hybrid bonding and data center demand[27].

Thermocompression bonders use heat and force to press memory dies into a stack and to attach chiplets to substrates. ASMPT says it remained the leader in that market in 2025 and aims for a 35 to 40 percent share[28]. In its 2025 results it said it expects the market to reach $1.6 billion by 2028[28]. In July 2026 ASMPT announced new bulk orders for more than 50 of its chip-to-substrate bonders for AI chips, from outsourced assembly and test firms[29]. Hanmi Semiconductor in South Korea also sells thermocompression bonders for high-bandwidth memory[30]. See Memory and HBM.

6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
16 Aehr Test Systems Reports Fiscal 2026 Fourth Quarter and Full Year Financial... Aehr Test Systems, 14 July 2026. aehr.com
20 Technoprobe, press release: the Board of Directors approves the draft statutory... Technoprobe, 18 March 2026, p. 1. technoprobe.com
21 Tokyo Seimitsu Integrated Report 2025. Accretech, 3 December 2025, p. 6. accretech.com
22 Precio™ Series. Tokyo Electron, accessed 9 September 2026. tel.com
23 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 20 October 2026. chipexplorer.eto.tech
24 COHU, INC., Form 10-K annual report for the period ended 2025-12-27 (10-K). U.S. Securities and Exchange Commission, 17 February 2026. sec.gov
25 DISCO develops the DFD6080 fully automatic dicing saw for work sizes up to 400... DISCO Corporation, 15 December 2025. disco.co.jp
26 KULICKE AND SOFFA INDUSTRIES, INC., Form 8-K, exhibit 99.1: results for the... U.S. Securities and Exchange Commission, 19 November 2025. sec.gov
27 BE Semiconductor Industries N.V. Announces Q2-26 and H1-26 Results. BE Semiconductor Industries (Besi), 23 July 2026. besi.com
28 ASMPT Announces 2025 Annual Results AI-Driven Structural Growth Underpins Group... ASMPT, 4 March 2026. asmpt.com
29 ASMPT Secures Significant Wins for More than 50 Chip-to-Substrate... ASMPT, 29 July 2026. asmpt.com
30 Hanmi Semiconductor, products page. Hanmi Semiconductor, accessed 9 September 2026. hanmisemi.com

04

The back end's geography

Taiwanese firms lead outsourced assembly and test, the OSAT business, and Chinese firms are large challengers. Taiwan's ASE billed $20,573.4 million in 2025[7]. Packaging made up about 48 percent of that, testing about 11 percent and its electronics manufacturing arm about 40 percent[31]. In a TrendForce ranking that CSET cites, four of the top ten OSATs by 2024 revenue are headquartered in China[9].

Malaysia is a large center for chip assembly and test. Its investment agency said in July 2026 that the country ships 13 percent of the world's packaged chips[32]. It said a consortium of five Malaysian firms aims to help the country capture 7 percent of world advanced packaging shipments by 2035[32]. A 2024 study by the Semiconductor Industry Association and Boston Consulting Group used a different measure, capacity[33]. It gave Malaysia 7 percent of global assembly, test and packaging capacity in 2022, the largest share in Southeast Asia[33]. The study measured capacity by factory floor space at contractors and chipmakers' own plants[34].

Taiwan's assembly and test contractors are expanding. ASE ran 6,797 testers in the second quarter of 2025 and 8,348 a year later[4]. It spent $2.7 billion on machinery and $1.4 billion on buildings, facilities and automation in the first half of 2026[35]. In July 2026 it said revenue from leading-edge advanced packaging was running ahead of its $3.5 billion forecast for 2026[35]. Powertech, another Taiwanese assembly and test firm, aims to run the first panel-level packaging line for AI chips in 2027[36].

Amkor, an American assembly and test contractor, took $6.71 billion of revenue in 2025[5]. Its revenue in the second quarter of 2026 alone was $1.90 billion, up 26 percent year on year[37]. By October 2025 it had raised the planned investment in its Arizona campus to $7 billion[10].

In October 2024 TSMC and Amkor signed a memorandum of understanding for TSMC to buy packaging and test services from Amkor's planned plant in

4 ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for... U.S. Securities and Exchange Commission, 30 July 2026. sec.gov

5 AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-02-09... U.S. Securities and Exchange Commission, 9 February 2026. sec.gov

7 ASE Technology Holding Co., Ltd., Form 20-F annual report for the period ended... U.S. Securities and Exchange Commission, 1 April 2026. sec.gov

9 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

10 Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test... Amkor Technology, 6 October 2025. ir.amkor.com

31 ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for... U.S. Securities and Exchange Commission, 5 February 2026. sec.gov

32 Advanced Packaging: How Malaysia is Packaging the Future of AI. Malaysian Investment Development Authority, 1 July 2026. mida.gov.my

33 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 21. semiconductors.org

34 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 35. semiconductors.org

35 ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for... U.S. Securities and Exchange Commission, 30 July 2026. sec.gov

36 Powertech eyes world's first panel-level packaging for AI chips in 2027. Nikkei Asia, 27 August 2026. asia.nikkei.com

37 AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-07-27... U.S. Securities and Exchange Commission, 27 July 2026. sec.gov

Peoria, Arizona[38]. TSMC would use those services for its customers, especially those using its Phoenix fabs[38]. The two announced a long-term partnership in June 2026[11].

Figure 13.4

ASE ran 8,348 testers in mid-2026, up from 6,797 a year earlier

Testers installed at ASE, a Taiwanese assembly and test contractor



Source ASE Technology Holding, second quarter 2026 earnings release (SEC Form 6-K) [4]

05

The chokepoint

Customers can switch tester makers, but at a cost. Advantest says that switching to another tester maker means rebuilding the whole test setup used for development, evaluation and mass production, at significant extra cost to the customer[8]. Still, ASE says it buys testers with similar functions from different makers, including Teradyne and Advantest[7].

Chinese toolmakers are gaining share in test and advanced packaging tools, but CSET finds they still lag significantly behind foreign suppliers in market share[9]. Between 2019 and 2024 they gained 5 percentage points of world market share in test tools and 4 points in fabrication tools built for advanced packaging, by CSET's analysis of TechInsights data[9]. But they lost share in bonding and other conventional assembly and packaging tools[9].

Assembly and test contractors such as ASE and Amkor earn a small part of what the chip industry sells. ASE's packaging business took $9,829.2 million from outside customers in 2025, and its testing business $2,292.0 million[7]. Amkor's revenue was $6.71 billion[5]. Global semiconductor sales that year were $791.7 billion[39].

Three questions to answer from memory

1. Why do chipmakers test chips early and often? see How it works
2. Who makes most chip testers? see Who makes it

4 ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for... U.S. Securities and Exchange Commission, 30 July 2026. sec.gov

5 AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-02-09... U.S. Securities and Exchange Commission, 9 February 2026. sec.gov

7 ASE Technology Holding Co., Ltd., Form 20-F annual report for the period ended... U.S. Securities and Exchange Commission, 1 April 2026. sec.gov

8 Investors Guide July 29, 2026. Advantest, 30 July 2026, p. 12. advantest.com

9 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

11 TSMC and Amkor Technology Announce Long Term Partnership to Accelerate Advanced... Amkor Technology, 16 June 2026. ir.amkor.com

38 Amkor and TSMC to Expand Partnership and Collaborate on Advanced Packaging in... TSMC, 4 October 2024. pr.tsmc.com

39 Global Annual Semiconductor Sales Increase 25.6% to $791.7 Billion in 2025. Semiconductor Industry Association, 6 February 2026. semiconductors.org

3. How many of the ten largest chip assembly and test contractors are based in China? see The back end's geography

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Three things to remember

- Chips are tested early and often because one bad chip can ruin a package of expensive ones.
- Advantest and Teradyne account for most spending on chip testers.
- In a TrendForce ranking cited by CSET, four of the top ten outsourced assembly and test firms by 2024 revenue are based in China.

Sources (39)

1. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 20 October 2026
   chipexplorer.eto.tech/?parentNode=N78&selectedNode=N120
2. A Advantest, presentation notes for the FY2025 (year ended 31 March 2026) results briefing, 27 April 2026. Advantest · 27 April 2026 · p. 13
   www.advantest.com/document/en/investors/ir-library/result/JE_BIZ_260427_note.pdf#page=13
3. A TERADYNE, INC, Form 10-K annual report for the period ended 2025-12-31 (10-K). U.S. Securities and Exchange Commission (filing by TERADYNE, INC) · 19 February 2026
   www.sec.gov/Archives/edgar/data/97210/000119312526059002/ter-20251231.htm
4. A ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for the period ended 2026-07-30. U.S. Securities and Exchange Commission (filing by ASE Technology Holding Co., Ltd.) · 30 July 2026
   www.sec.gov/Archives/edgar/data/1122411/000095010326011351/dp250868_6k.htm
5. A AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-02-09 (8-K). U.S. Securities and Exchange Commission (filing by AMKOR TECHNOLOGY, INC.) · 9 February 2026
   www.sec.gov/Archives/edgar/data/1047127/000104712726000007/amkr123125erex-991.htm
6. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 9 September 2026
   chipexplorer.eto.tech/?parentNode=N69&selectedNode=N71
7. A ASE Technology Holding Co., Ltd., Form 20-F annual report for the period ended 2025-12-31. U.S. Securities and Exchange Commission (filing by ASE Technology Holding Co., Ltd.) · 1 April 2026
   www.sec.gov/Archives/edgar/data/1122411/000119312526135585/d50802d20f.htm
8. A Investors Guide July 29, 2026. Advantest · 30 July 2026 · p. 12
   www.advantest.com/document/en/investors/ir-library/investors-guide/Investors_Guide_2607E.pdf#page=12
9. A Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology (CSET) · 14 July 2025
   cset.georgetown.edu/article/inside-beijings-chipmaking-offensive/
10. A Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test Campus in Arizona; Expands Investment to $7 Billion. Amkor Technology · 6 October 2025
    ir.amkor.com/news-releases/news-release-details/amkor-technology-breaks-ground-new-semiconductor-advanced
11. A TSMC and Amkor Technology Announce Long Term Partnership to Accelerate Advanced Packaging in the United States. Amkor Technology · 16 June 2026
    ir.amkor.com/news-releases/news-release-details/tsmc-and-amkor-technology-announce-long-term-partnership
12. A Testimony of David Isaacs, Semiconductor Industry Association, hearing on "Trade in Critical Supply Chains". U.S. Senate Committee on Finance · 14 May 2025 · p. 2
    www.finance.senate.gov/download/051425-isaacs-testimony&download=1#page=2
13. A Investors Guide July 29, 2026. Advantest · 30 July 2026 · p. 10
    www.advantest.com/document/en/investors/ir-library/investors-guide/Investors_Guide_2607E.pdf#page=10
14. A AEHR TEST SYSTEMS, Form 10-K annual report for the period ended 2026-05-29. U.S. Securities and Exchange Commission (filing by AEHR TEST SYSTEMS) · 27 July 2026
    www.sec.gov/Archives/edgar/data/1040470/000165495426006919/aehr_10k.htm
15. A Aehr Receives $22 Million Follow-On Order for AI Processor Wafer-Level Burn-In Systems. Aehr Test Systems · 12 August 2026
    www.aehr.com/2026/08/aehr-receives-22-million-follow-on-order-for-ai-processor-wafer-level-burn-in-systems/
16. A Aehr Test Systems Reports Fiscal 2026 Fourth Quarter and Full Year Financial Results with Record Quarterly Bookings and $100 Million Effective Backlog. Aehr Test Systems · 14 July 2026
    www.aehr.com/2026/07/aehr-test-systems-reports-fiscal-2026-fourth-quarter-and-full-year-financial-results-with-record-quarterly-bookings-and-100-million-effective-backlog/
17. A FORMFACTOR INC, Form 10-K annual report for the period ended 2025-12-27 (10-K). U.S. Securities and Exchange Commission (filing by FORMFACTOR INC) · 20 February 2026
    www.sec.gov/Archives/edgar/data/1039399/000103939926000009/form-20251227.htm

18. A Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese, English) — edited transcript. LSEG StreetEvents (transcript of a TSMC earnings call), via TSMC · 16 July 2026 · p. 16
investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2026-08/3e494f0c14dd0890f897aa044415e21d93486cc4/TSMC%202Q26%20Transcript.pdf#page=16

19. A Teradyne Reports Fourth Quarter and Full Year 2025 Results. Teradyne · 2 February 2026
investors.teradyne.com/news-events/press-releases/detail/433/teradyne-reports-fourth-quarter-and-full-year-2025-results

20. A Technoprobe, press release: the Board of Directors approves the draft statutory and consolidated annual report as at 31 December 2025. Technoprobe · 18 March 2026 · p. 1
technoprobe.com/wp-content/uploads/2026/03/PR-FY-2025_.pdf#page=1

21. A Tokyo Seimitsu Integrated Report 2025. Accretech · 3 December 2025 · p. 6
www.accretech.com/en/ir/index/main/00/teaserItems1/00/linkList/02/link/IntegratedReport2025_E.pdf#page=6

22. A Precio™ Series. Tokyo Electron · Accessed 9 September 2026
www.tel.com/product/precio.html

23. A Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) · Accessed 20 October 2026
chipexplorer.eto.tech/?parentNode=N78&selectedNode=N83

24. A COHU, INC., Form 10-K annual report for the period ended 2025-12-27 (10-K). U.S. Securities and Exchange Commission (filing by COHU, INC.) · 17 February 2026
www.sec.gov/Archives/edgar/data/21535/000143774926004339/cohu20251227_10k.htm

25. A DISCO develops the DFD6080 fully automatic dicing saw for work sizes up to 400 mm square (最大400mm角のパッケージ切断に対応したダイシングソー「DFD6080」を開発). DISCO Corporation · 15 December 2025
www.disco.co.jp/eg/news/corp/20251215_1.html

26. A KULICKE AND SOFFA INDUSTRIES, INC., Form 8-K, exhibit 99.1: results for the fourth quarter and fiscal 2025 (8-K). U.S. Securities and Exchange Commission (filing by KULICKE AND SOFFA INDUSTRIES, INC.) · 19 November 2025
www.sec.gov/Archives/edgar/data/56978/000005697825000078/ex991liveq42025.htm

27. A BE Semiconductor Industries N.V. Announces Q2-26 and H1-26 Results. BE Semiconductor Industries (Besi) · 23 July 2026
www.besi.com/investor-relations/press-releases/2025/details-1/be-semiconductor-industries-nv-announces-q2-26-and-h1-26-results/

28. A ASMPT Announces 2025 Annual Results AI-Driven Structural Growth Underpins Group Performance. ASMPT · 4 March 2026
www.asmpt.com/en/investor-relations/news-events/asmpt-announces-2025-annual-results/

29. A ASMPT Secures Significant Wins for More than 50 Chip-to-Substrate Thermo-Compression Bonding Tools. ASMPT · 29 July 2026
www.asmpt.com/en/investor-relations/news-events/asmpt-secures-significant-wins/

30. A Hanmi Semiconductor, products page. Hanmi Semiconductor · Accessed 9 September 2026
www.hanmisemi.com/en/product/index.html

31. A ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for the period ended 2026-02-05. U.S. Securities and Exchange Commission (filing by ASE Technology Holding Co., Ltd.) · 5 February 2026
www.sec.gov/Archives/edgar/data/1122411/000095010326001661/dp241159_6k.htm

32. A Advanced Packaging: How Malaysia is Packaging the Future of AI. Malaysian Investment Development Authority · 1 July 2026
www.mida.gov.my/advanced-packaging-how-malaysia-is-packaging-the-future-of-ai/

33. A Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry Association · May 2024 · p. 21
www.semiconductors.org/wp-content/uploads/2024/05/Report_Emerging-Resilience-in-the-Semiconductor-Supply-Chain.pdf#page=21

34. A Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry Association · May 2024 · p. 35
www.semiconductors.org/wp-content/uploads/2024/05/Report_Emerging-Resilience-in-the-Semiconductor-Supply-Chain.pdf#page=35

35. A ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for the period ended 2026-07-30. U.S. Securities and Exchange Commission (filing by ASE Technology Holding Co., Ltd.) · 30 July 2026
www.sec.gov/Archives/edgar/data/1122411/000095010326011353/dp250875_6k.htm

36. B Powertech eyes world's first panel-level packaging for AI chips in 2027. Nikkei Asia · 27 August 2026
asia.nikkei.com/business/tech/semiconductors/powertech-eyes-world-s-first-panel-level-packaging-for-ai-chips-in-2027

37. A AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-07-27 (8-K). U.S. Securities and Exchange Commission (filing by AMKOR TECHNOLOGY, INC.) · 27 July 2026
www.sec.gov/Archives/edgar/data/1047127/000104712726000043/amkr6302026erex-991.htm

38. A Amkor and TSMC to Expand Partnership and Collaborate on Advanced Packaging in Arizona. TSMC · 4 October 2024
pr.tsmc.com/english/news/3174

39. A Global Annual Semiconductor Sales Increase 25.6% to $791.7 Billion in 2025. Semiconductor Industry Association · 6 February 2026
www.semiconductors.org/global-annual-semiconductor-sales-increase-25-6-to-791-7-billion-in-2025/

Made by **Yashvardhan Sharma** with Claude. Corrections and questions: contact@chipsupplychain.org.

A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

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